BLACK PEN Answer all questions first. **GREEN PEN** Then tick correct answers; rewrite mistakes.

CPU Performance: Clock Speed, Cache and Cores

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: Answer all four before looking at the model answers.

1. What does MAR store?

2. Name the CPU component that performs arithmetic.

3. What does the PC store?

4. Describe a fetch using PC, MAR and MDR.

Today's objectives

Explain how clock speed, cache and cores affect CPU performance. Compare processors using clear evidence.

Key vocabulary - Read, cover, recall

clock speed
Number of CPU clock cycles per second.

cache
Small, fast memory holding frequently used information.

core
CPU processing unit able to execute instructions.

parallel processing
Running work simultaneously using multiple processing units.

Write one definition in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then tick correct answers; rewrite mistakes.

Learn and practise

New learning - 1.1.2

Clock speed	Clock speed is measured in Hz: clock cycles per second. One cycle does not necessarily equal one instruction.
Cache size	Cache stores frequently needed information closer to the CPU, reducing time spent accessing RAM.
Core count	Multiple cores enable more work in parallel when software supports simultaneous tasks.
Overall comparison	Architecture and task type also affect performance; do not infer an absolute winner from one specification.

Which processor is faster? Compare the workload.

CPU A

2 GHz | 4 cores

4 MB cache

CPU B

3 GHz | 2 cores

8 MB cache

Clock, core count and cache favour different workloads.

I Do - Worked example

Compare CPU A (2 GHz, 4 cores, 4 MB cache) with B (3 GHz, 2 cores, 8 MB cache). Which is faster?

1. CPU B has the higher clock speed.
2. CPU A has more cores, which can help with suitably parallel work.
3. CPU B has more cache, potentially reducing RAM access.
4. Explain that workload and architecture determine real performance.
5. Conclude neither is automatically fastest for every task.

We Do - Guided practice

CPU C is 2 GHz and CPU D is 3 GHz. Identify which has a higher clock speed.

Use the model. Write your answer in BLACK PEN:

BLACK PEN Answer all questions first. **GREEN PEN** Then tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

Read the code before you write or test anything.

```
clockGHz = 3
seconds = 2
print(clockGHz * seconds)
```

Predict the number of billions of clock cycles in two seconds.

1. Predict the number of billion cycles.

2. Change seconds to four.

3. Explain why cycle count is not the number of executed instructions.

You Do - Independent exam questions

Q1. State two characteristics affecting CPU performance. [2 marks]

Q2. Explain how a larger cache may improve performance. [2 marks]

Q3. Two CPUs both run at 2 GHz but one has twice as many cores. Explain why it may not always finish a task faster. [4 marks]

Exit Ticket - True or False?

Answer each statement before turning to the answers.

1. 3 GHz guarantees exactly three billion instructions every second. ☐ T ☐ F

2. A larger cache may reduce time spent accessing RAM. ☐ T ☐ F

3. Twice the cores always doubles a single-threaded program's speed. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick what is correct, then use the model to fix mistakes.
Do not just copy answers; explain what you changed.

Do Now - Answers

Q1 The accessed memory address. ☐

Q2 ALU. ☐

Q3 Address of the next instruction. ☐

Q4 PC supplies next instruction address; MAR holds accessed address; MDR holds fetched instruction/data. ☐

Code First - Model

6 billion cycles; then 12 billion. An instruction can require different numbers of clock cycles and architecture matters.

We Do - Model answer

CPU D.

Exit Ticket - Check

1. FALSE - False: revise the new learning.
2. TRUE - True: useful data can be available closer to the CPU.
3. FALSE - False: a single thread may not use extra cores.

You Do - Self-mark your exam answers

Q1 Any two from clock speed, cache size, number of cores [1 each]. ☐

Q2 Holds frequently used data/instructions near CPU [1], reducing slower RAM accesses [1]. ☐

Q3 More parallel capacity [1]; only when software uses multiple threads [1]; single-threaded code cannot fully use extra cores [1]; other factors such as cache and architecture matter [1]. ☐

Try the question again

1. Copy the FULL question you got wrong (or found hardest):

2. In GREEN PEN, write a NEW complete answer. Show working: